

PROJECT REPORT

SmartServe AI: Demand Forecasting & Waste Reduction

Hackathon Submission - Machine Learning Domain

Project Report: SmartServe AI

Food Demand Prediction and Waste Reduction System

Hackathon Domain: Machine Learning

Project Title: SmartServe AI

Deployment Status: Live on Streamlit Community Cloud

Repository: github.com/sadaf2730/smartserve-ai

1. Executive Summary

SmartServe AI is an end-to-end Machine Learning web application designed to solve the dual challenges of food waste and meal shortages in large-scale catering services (college canteens, hostels, cafeterias, and event venues).

Traditionally, kitchen managers rely on guesswork, over-preparing by **15% to 25%** to avoid stockouts. SmartServe AI replaces this heuristic with a **Random Forest Regressor** that forecasts precise food demand **BEFORE** preparation begins. By integrating historical demand variability (sigma (variability)), the system recommends an **Optimal Preparation Quantity** that minimizes both leftovers (waste) and shortages (hungry guests).

On a 6-month historical dataset (540 records), the predictive engine achieved an **R^2 accuracy score of 0.9762** and a **Mean Absolute Error (MAE) of 6.38 servings**, demonstrating a dynamic waste reduction potential of **over 45%** with zero customer shortage risk.

[Problem] 2. Problem Statement

Food service operations suffer from severe inefficiencies due to demand uncertainty:

- 1. The Overproduction Cost (Food Waste):** Canteens routinely throw away 15% of cooked food. Landfilled food waste decomposes anaerobically, generating methane-a greenhouse gas 25x more potent than carbon dioxide. Financially, this discards thousands of dollars in raw ingredients and labor daily.

2. **The Underproduction Cost (Shortage):** Preparing too little food leads to stockouts, resulting in student/customer dissatisfaction, long queues, and nutritional deficits.

3. **The Guesswork Loop:** Attendance is highly volatile, shifting due to weather changes, weekday/weekend schedules, calendar holidays, and special campus events. Catering managers lack tools to extract predictive patterns from these complex, overlapping variables.

[Solution] 3. Proposed Solution

SmartServe AI provides an intelligent, automated dashboard to manage and optimize food preparation:

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Historical Food Data

Data Cleaning & Feature Engineering

Train/Test Split (80/20)

Random Forest Regression (Demand Forecast)

Variability Analysis (Optimal Serving Bounds)

Dynamic Risk Profiling (Shortage vs. Waste Risk)

Interactive Streamlit Dashboard & Simulator

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- **Target Leakage Prevention:** We strictly isolate variables available *before* cooking. The model predicts actual customer demand (`quantity\_consumed`) without using the current service's prepared limits as a feature.
- **Variability-Led Optimization:** Instead of static safety buffers, the system dynamically calculates standard deviations (sigma (variability)) for each meal type. It provides kitchen staff with an explainable recommendation matching the predicted demand plus a variability-scaled buffer, protecting against service

spikes.

- **Shortage & Waste Risk Profiling:** Classifies operating plans into Low/Medium/High risk categories so managers understand the visual trade-offs of their cooking decisions in real-time.

4. Data Layer & Feature Engineering

The system trains on a 6-month simulated sequential dataset (540 records) representing three daily services (Breakfast, Lunch, Dinner). The dataset incorporates realistic relationships, weekend patterns (e.g., student sleeping habits), and seasonal transitions:

Training Features (Pre-Preparation Variables):

- **Expected Customer Registrations:** RSVP counts or ticket bookings.
- **Meal Type:** Breakfast, Lunch, or Dinner (distinct consumption profiles).
- **Day of Week & Weekend Indicator:** Monday-Sunday classifications capturing weekend attendance drops.
- **Weather & Temperature:** Simulates weather fluctuations (e.g., rainfall increases dining hall traffic).
- **Special Event Indicator:** Binary flag representing student festivals or campus holidays.
- **Temporal Lag Features:**
 - ``prev_quantity_consumed`` & ``prev_quantity_wasted`` (captures immediate demand swings).
 - ``rolling_avg_consumption`` (3-meal rolling average of the specific meal type).
 - ``recent_demand_trend`` (momentum difference relative to the rolling average).

5. Machine Learning & Model Performance

We trained a **Random Forest Regressor** (100 estimators) and compared it against a baseline **Linear Regression** model using an 80/20 train/test split.

Performance Metrics (On Test Split):

Metric	Random Forest (SmartServe AI)	Linear Regression (Baseline)
Mean Absolute Error (MAE)	6.38 servings	7.95 servings
Root Mean Squared Error (RMSE)	8.14 servings	10.12 servings
Coefficient of Determination (R^2)	0.9762	0.9632

- **Feature Importance:** Random Forest diagnostics reveal that **Expected Customers** holds the highest predictive weight (~68%), followed closely by **Rolling Average Consumption** (~18%) and **Meal Type** (~8%), proving that temporal lag features are vital for forecasting.
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6. Optimal Preparation & Risk Logic

1. Optimal Preparation Quantity

To calculate recommended preparation servings (Q_{opt}), the prediction pipeline uses predicted demand (Predicted Demand), the meal's historical standard deviation (σ (variability)), and a configurable safety factor (Safety Factor, defaults to 0.8):

$$Q_{opt} = \text{ceil}(\text{Predicted Demand} + (\text{Safety Factor} * \sigma_{\text{meal}}))$$

If a special event is flagged, the safety factor is bumped by \$+0.30\$ to prevent stockouts during volatile guest spikes.

2. Dynamic Risk Classifications

For any target preparation quantity Prepared Quantity chosen by a chef:

- **Shortage Risk:**
 - **High:** $Q < D_{\text{pred}}$ (guaranteed stockout)
 - **Medium:** $D_{\text{pred}} \leq Q < D_{\text{pred}} + 0.5\sigma_{\text{meal}}$ (vulnerable to unexpected customer spikes)
 - **Low:** $Q \geq D_{\text{pred}} + 0.5\sigma_{\text{meal}}$ (safe buffer zone)
 - **Waste Risk:**
 - **Low:** $Q \leq D_{\text{pred}} + 0.5\sigma_{\text{meal}}$ (minimal leftovers)
 - **Medium:** $D_{\text{pred}} + 0.5\sigma_{\text{meal}} < Q \leq D_{\text{pred}} + 1.5\sigma_{\text{meal}}$ (manageable waste)
 - **High:** $Q > D_{\text{pred}} + 1.5\sigma_{\text{meal}}$ (excessive overproduction)
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7. Business & Ecological Impact

The landing page features a live **Waste Reduction Impact** panel that dynamically simulates performance over the 6-month historical logs:

- **Baseline Historical Waste Rate: 13.9%** (overproduction due to guesswork).
 - **AI-Optimized Waste Rate: 7.2%** (wasted servings restricted to the optimal safety margin).
 - **Potential Waste Reduction: ~48.2% less food waste.**
 - **Wasted Servings Rescued: 3,524 servings** saved.
 - **Estimated Financial Savings: \$8,810.00** saved (assuming \$2.50 per serving ingredient cost).
 - **Carbon Offset equivalent: 3,524 kg CO2 equivalent** prevented from entering landfills (assuming 1 serving = 0.4 kg food; 1 kg waste = 2.5 kg CO2).
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8. Future Scope

1. **Recipe Scaler Integration:** Automate back-of-house kitchen prep by converting recommended servings directly into exact ingredient weights (kilograms of rice, vegetables, meat).
2. **IoT Waste Bins Integration:** Connect with smart load cell scales placed on kitchen trash cans to automatically feed actual wastage metrics back into the system.
3. **Student RFID Portal Sync:** Link directly with campus card swipes and lecture timetables to gain real-time, hour-by-hour customer expectation signals.